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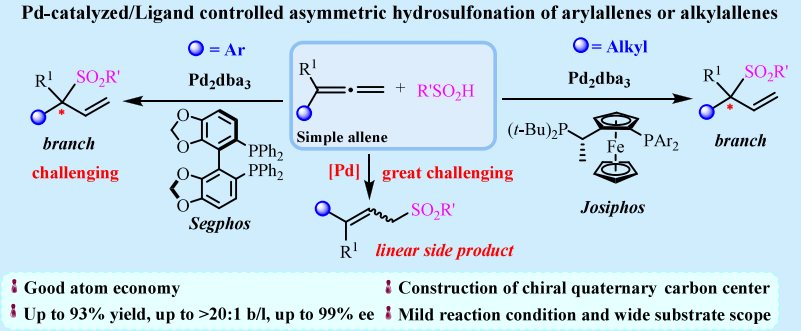
Pd-Catalyzed/Ligand-Controlled Regioselective Asymmetric Hydrosulfonylation of Alkylallenes or Arylallenes

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ABSTRACT: The metal-catalyzed hydrofunctionalization reaction of allenes is an efficient approach for the construction of new allyl compounds. This work described a palladium-catalyzed/ligand-controlled asymmetric hydrosulfonylation of simple alkylallenes and arylallenes in the presence of Pd_2dba_3. Using a Josiphos or Segphos ligand, the alkylallenes or arylallenes could render the corresponding branched chiral allylsulfones with high yields and high enantioselectivities up to 90% and 99% ee, respectively.

The significance of allene is self-evident in organic chemistry because of its unique linear three-carbon skeleton constructed by two cumulative π bonds, thus endowing its interesting reactivities and diverse site selectivities (α-, β-, or γ-).[1] Many reviews have emerged on their preparation and transformation.[2] Among them, the transition-metal-catalyzed asymmetric hydrofunctionalization of allenes represents an important domain in allene chemistry. It is a powerful means to construct allyl C−C and C−X bonds with chiral carbon centers from allenes, and it can also directly give rise to many natural and non-natural products as well as some important pharmaceutical molecules.[3] In these reactions, Rh and Ir catalysts are commonly used to achieve the formation of chiral branched products, but most need a higher temperature or longer reaction time.[4] As a comparison, Pd-catalyzed asymmetric functionalization reactions of allenes have been extremely scarce until now.[5] In addition, Pd-catalyzed hydrofunctionalization of allenes usually results in a dominated linear allyl product (Scheme 1a).[6] And compared with the alkylallenes, the arylallenes are relatively difficult substrates and more inclined to render the linear products without branched products, as may be due to electronic effects and conjugated effects as well as possible steric effects of the aromatic ring on the arylallenes.[7] Therefore, it is highly desirable to regulate the branched regioselectivity of the asymmetric hydrofunctionalization of allenes precisely and remains greatly challenging, especially for the arylallenes.

Considering that, we herein focus on the asymmetric hydrosulfonylation of simple terminal allenes including alkyl- and arylallenes, with the aim of the synthesis of diverse chiral allylsulfones, which are the ubiquitous building blocks of the drugs and natural products (Scheme 1b).[8] Breit and his co-workers developed a Rh-catalyzed asymmetric hydrothioetheration oxidation of allenes in a two-step process to afford the chiral allylsulfones. In their reaction, excessive oxidants and smelly sulfur-containing toxic reagents were used and only limited to alkylallenes (Scheme 1c, Breit's work).[9] Recently, Aponick and co-workers reported an asymmetric hydrosulfonylation reaction of allenes through using a Rh catalyst (Scheme 1c, Aponick's work).[10] In their work, the arylallenes delivered the corresponding allyl sulfones with relatively low enantioselectivities. In spite of this, we herein developed a Pd-catalyzed/ligand-controlled asymmetric hydrosulfonylation of alkylallenes or arylallenes, and a series of chiral allylsulfones

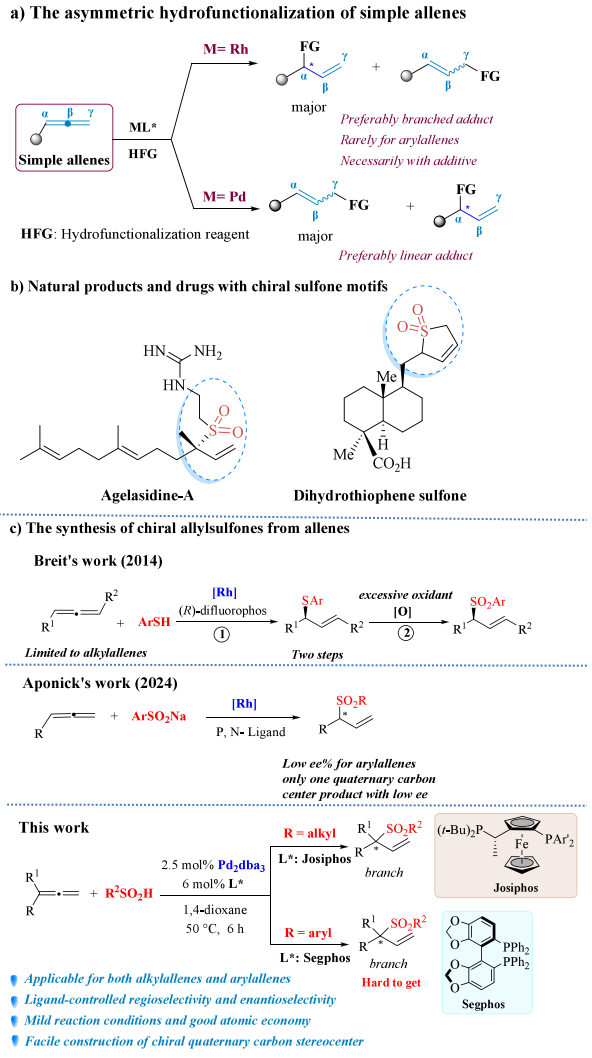
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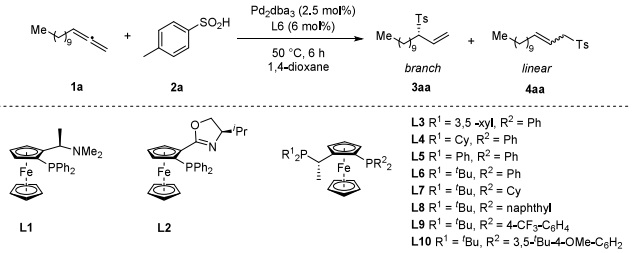
Scheme 1. Synthesis and Application of Chiral Allylsulfones from Allenes



were obtained in good yields with excellent enantioselectivities. Namely, using a Pd_2dba_3/Josiphos or Pd_2dba_3/Segphos chiral metal complex, the alkylallenes or arylallenes could smoothly proceed to give the corresponding branched chiral allylsulfones, respectively (Scheme 1c, this work). This reaction breaks through the inherent reaction mode of palladium catalysis with dominant linear products, especially for the challenging arylallenes. Moreover, this methodology is also applicable for the construction of allylsulfones containing chiral quaternary carbon stereocenters. And the DFT calculations also consolidated the ligand-controlled regioselectivity and enantioselectivity of the asymmetric hydrosulfonylation reaction.

Our initial studies commenced with the hydrosulfonylation of the trideca-1,2-diene **1a** and *p*-toluenesulfinic acid **2a** in the presence of Pd_2dba_3 catalyst for screening of various chiral phosphine ligands (**L1**−**L10**) with the aim of identifying conditions for a direct catalytic enantioselective reaction (Table 1). After an extensive examination, it was found that the substituents on Josiphos had a significant effect on the reaction enantioselectivities (Table 1, entries 1−12), and the ligand **L6** (R^1 = *tert*-butyl, R^2 = Ph) performed the best, rendering the corresponding allylsulfone branched product **3aa**

Table 1. Optimization of the Chiral Ligands for the Reaction between Alkylallene 1a and *p*-Tolsulfinic Acid 2a[a]



Entry	Variants from the standard condition	Yield of **3aa** (%)[b]	*b*/*l*[b]	ee (%)[c]
1	none	93	>95:5	86
2	**L1**	n. r.		
3	**L2**	n. r.	-	-
4	**L3**	45	93:7	12
5	**L4**	18	93:7	47
6	**L5**	81	92:8	54
7	**L6**	**93**	**>95:5**	**86**
8	**Enti-L6**	81	>95:5	81
9	**L7**	40	88:12	58
10	**L8**	70	95:5	41
11	**L9**	88	94:6	85
12	**L10**	64	88:12	58
13	THF	27	50:50	31
14	$Pd(dba)_2$	43	91:9	80
15	$Pd(PPh_3)_4$	26	-	17
16	[Pd]:**L6**=1:1	38	50:50	26

[a]Reaction procedure: The trideca-1,2-diene **1a** (18 mg, 0.1 mmol), *p*-tolsulfinic acid **2a** (18.7 mg, 0.12 mmol, 1.2 equiv), Pd_2dba_3 (0.0025 mmol, 0.025 equiv), ligand (0.006 mmol, 0.06 equiv), and 1,4-dioxane were sequentially added into a Schlenk tube, and the mixture was stirred at 50 °C (an oil bath) for 6 h under an argon atmosphere. [b]Yields and *b*/*l* ratio were determined by ^{1}H NMR spectroscopy of crude product using 4-bromobenzaldehyde as an internal standard sample. [c]Determined by chiral HPLC. n. r. = no reaction.

in high up to 93% yield and 86% ee as well as a *b*/*l* ratio > 95:5 (Table 1, entry 6). Replacing the 1,4-dioxane with THF, we found that the yield and ee value of the corresponding product **3aa** decreased rapidly (entry 13). Other Pd catalysts, such as $Pd(dba)_2$ and $Pd(PPh_3)_4$, both showed poor performance (entries 14 and 15). And the ratio of Pd/**L6** was also examined (entry 16). Other details of experimental parameters on the reaction of **1a** and **2a**, such as the Pd catalyst, solvent, the ratio of Pd catalyst, ligand and additive, all were investigated and shown in Supporting Information (see Tables S1−S3). As a result, the optimized reaction conditions were determined to be that the alkylallenes **1** (0.1 mmol, 1.0 equiv) and sulfones **2** (1.2 equiv) were reacted in 1,4-dioxane (1.6 mL) at 50 °C for 6 h under an argon atmosphere with using Pd_2dba_3 (2.5 mol %) and Josiphos ligand **L6** (6 mol %).

In the case of arylallene **1s**, only a trace of branched **3sa** was obtained with low enantioselectivity under the above optimal reaction conditions (Table S4, entry 1), and the linear product **4sa** was dominant. This result might be ascribed to the smaller steric hindrance and greater stability of the large conjugated aromatic system for the linear product generated. Thus, other chiral ligands were further examined (entries 2−11). As a result, the high enantioselectivity of **3sa** was obtained in moderate yield (52% yield, 94% ee) with a 1.1/1 ratio of branched and linear products when using the phenyl-

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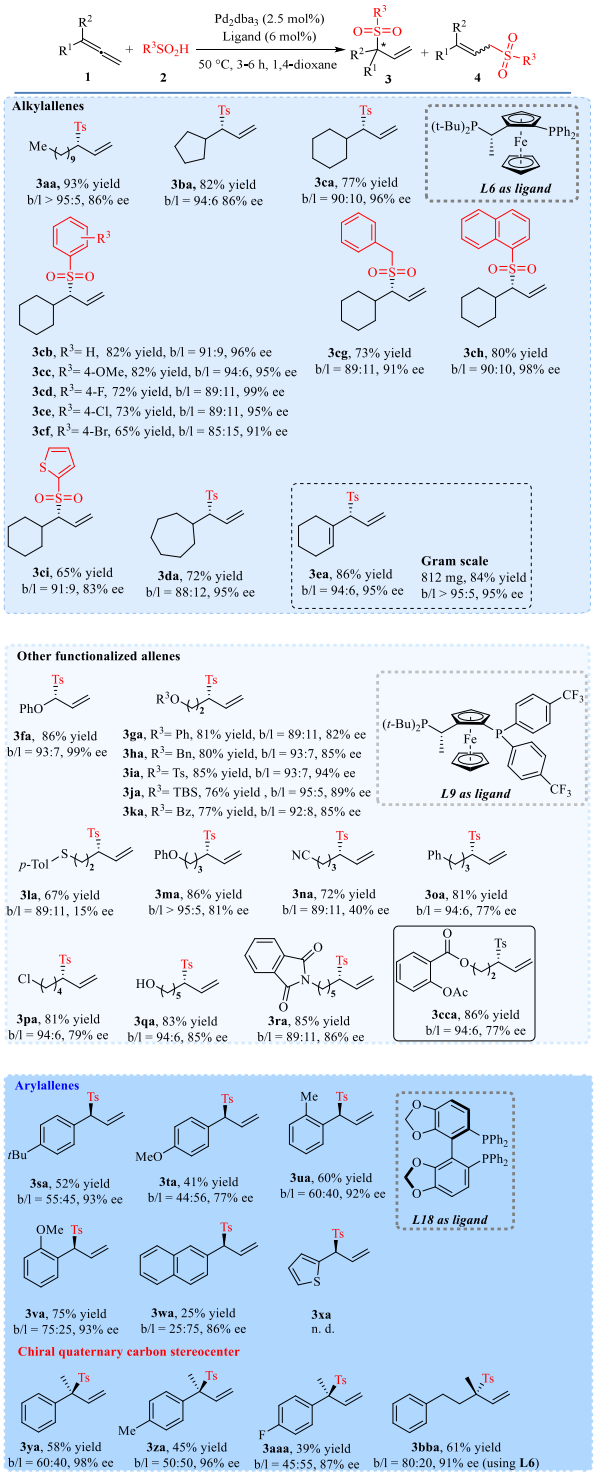
substituted Segphos ligand **L18** (Table S4, entry 9). In view of the poor *b*/*l* ratio, we herein also tried to use the Rh catalyst with some chiral ligands. The results are shown in Table S5. But consequently, no satisfying yield and ee value of **3sa** were observed, although the *b*/*l* ratio was good. Other experimental parameters were further screened using the Pd catalyst, and no obvious improved results were obtained (Table S6). Therefore, the optimal reaction conditions for the asymmetric hydrosulfonylation of arylallenes were determined to be similar to those of the alkylallenes except for using ligand **L18** at a short time for 3 h.

Under the above optimal reaction conditions, we evaluated substrate scope with different alkylallene or arylallene substrates **1** and sulfones **2**. As shown in Table 2, the alkylallenes, whether cyclohexyl or cyclopentyl as well as cycloheptyl, all aliphatic allenes could smoothly proceed and deliver the corresponding chiral allylsulfones in good yields and regioselectivities as well as high enantioselectivities (**3aa**−**3da**). In the case of cyclohexylallene **1c**, a series of sulfones **2** was also investigated, and it was found that various substituted arylsulfinic acids were good candidates for the asymmetric hydrosulfonylation (**3cb**−**3cf**). In detail, it was observed that the substituents of the arylsulfinic acid had little effect on this reaction. And the 4-OMe-substituted sulfinic acid gave the corresponding product with a high *b*/*l* ratio up to 18:1 of branched/linear regioselectivity (**3cc**), which might be because the electron-donating substituted sulfinic acid could stabilize the allyl palladium intermediate well. Moreover, the benzyl-, sterically hindered naphthyl-, and thiophenylsulfinic acid were also tolerant and gave the target molecules **3cg**−**3ci** in good yields and high enantioselectivities.

In addition, using chiral ligand **L9**, other substituted allenes, cyclohexenyl- and phenoxylallene, could also give satisfactory results with high yields, regioselectivities, and enantioselectivities (**3ea** and **3fa**). And the scale-up reaction of cyclohexenylallene could also smoothly proceed and afford **3ea** in 812 mg. In addition, a variety of substituted ethylallenes with PhO−, benzyloxy (BnO−), benzenesulfonyl (Ts−), tributylsilyl (TBS−), and benzoxyl (Bz−) groups on the terminal end of the ethyl chain produced the corresponding chiral allylsulfones **3ga**−**3ka**. Nevertheless, when O was replaced with a S atom, the enantioselectivity of the reaction product sharply decreased to 15% ee (**3la**). The reason might be attributed to the poisoning of the active chiral palladium species by sulfur atoms. The alkylallenes with a prolonged carbon chain, no matter what substituents are on the end of the chain (such as PhO−, −CN, Ph−, Cl−, or even −OH), all were still tolerant and rendered the corresponding chiral allylsulfones **3ma**−**3qa**. The phthalimide-substituted alkylallene **1r** was also tolerant and gave the corresponding product **3ra** in good yield and enantioselectivity. Moreover, the derivative of aspirin **1cc** could also be modified to produce the chiral allylsulfone **3cca** in 86% yield with 80% ee.

The reaction of the arylallenes and sulfinic acid was achieved through using sterically hindered ligand segphos (**L18**) with strong electron-donating ability and led to the branched product with high enantioselectivity up to 93% ee. And several substituted arylallenes could deliver the corresponding branched products **3sa**−**3va** with high enantioselectivities. But unfortunately, the regioselectivity of the branched and linear product was unsatisfying in the presence of the Pd catalyst. In spite of this, it was found that *ortho*-methoxyphenylallene could give the product with good *b*/*l*

Table 2. Evaluation of the Substrate Scope of Various Allenes 1 and Sulfinic Acids 2[a]



[a]Reaction procedure: The allene substrates **1** (0.1 mmol) and **2** (0.12 mmol), Pd_2dba_3 (0.0025 mmol), chiral ligand (0.006 mmol), and 1,4-dioxane were sequentially added into a Schlenk tube, and the mixture was stirred at 50 °C (an oil bath) under an argon atmosphere for 3 h (arylallenes) or 6 h (alkylallenes). [b]Yields and b/l value were determined by ^{1}H NMR spectroscopy of crude product using 4-bromobenzaldehyde as an internal standard sample. [c]Determined by chiral HPLC. Isolated yield. n. r. = no reaction. n. d. = no observed.

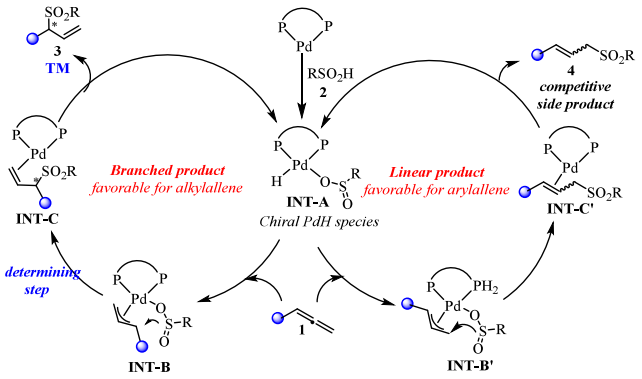
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regioselectivity (**3va**). From the results of **3ua** and **3va**, we could surmise that the oxygen atom of the *ortho*-methoxy group might participate in the coordination with the Pd complex, as it obviously improved the regioselectivity. In addition, the steric hindrance naphthylallene could also react with the phenylsulfinic acid to afford **3wa**, but thiophenylallene was not suitable for this asymmetric hydrosulfonylation reaction (**3xa**). Interestingly, 1,1-substituted arylallenes or alkylallene were also good candidates and rendered the corresponding allysulfone products containing chiral quaternary carbon stereocenters with high enantioselectivity (**3ya**, **3za**, **3aaa**, and **3bba**).

To explore the reaction mechanism, we performed the kinetic isotope effect (KIE) control experiments through using cyclohexylallene **1c** and 100% deuterated substrate **2a**−**d** (Scheme S1a). The K_H/K_D ratio of 1.8 demonstrated that this reaction was not first-order KIE and that the cleavage of the O−H bond was not the rate-determining step. And the 61% deuterated product **3aa**-*d* indicated that the hydrogen at the allyl site majorly came from sulfinic acid and minorly from a trace of water in the reaction system, solvent, or reactants (Scheme S1b). Combining these results with the absolute configuration of **3oa** and **3sa** determined by the single X-ray diffraction (see Figures S1 and S2), we proposed a possible transition-state model (Figure S3). In the system of (*S*, *Rp*)-Josiphos with allyl palladium, the sulfinic acid prefers the *Si* face instead of the *Re* face to attack and gives the **3oa**. In the chiral transition state formed by phenyl-substituted allenes with (*R*)-Segphos/palladium, sulfinic acid is also attacked from the less sterically hindered *Si* surface to obtain **3sa**.

Also, combining the reported results from relevant literature[11] with our dynamic isotope effect (KIE) experimental results (more details shown in the SI), we proposed a possible catalytic reaction mechanism as illustrated in Scheme 2. In the case of arylallene, the palladium catalyst initially

Scheme 2. Plausible catalytic cycle for the arylallenes



inserts into the O−H bond of RSO_2H to generate an M−H intermediate (**INT-A**). Subsequently, the allene compound undergoes a hydrogenation reaction, resulting in the formation of a π-allyl palladium complex (**INT-B**). Following this step, the lone pair of electrons on the sulfur atom attacks the π-allyl group while simultaneously cleaving the Pd−O bond to form a π-Pd complex (**INT-C**). Ultimately, **INT-C** dissociates from palladium, resulting in the formation of the final product and facilitating catalyst regeneration. And it was surmised that the regioselectivity- and enantioselectivity-determining step in this reaction may involve the transformation from **INT-B** to **INT-C**.

To further understand how the catalyst controls regioselectivity and stereoselectivity, we selected the Pd/**L18**-catalyzed reaction of arylallenes as a model for density functional theory (DFT) calculations, focusing on locating the transition state structure of this conversion process. We located the transition states corresponding to the generation of linear and branched products (*R*, *S*), and we calculated their relative energies. As shown in Figure S4 (for more less-favorable transition states, see the Supporting Information file), the transition state (TS-branched-*S*) for generating the *S*-branched product possesses the lowest energy, while the energy of its enantiomer product (TS-branched-*R*) is 2.6 kcal/mol higher. The computational results predict that the ee value of the branched product is approximately 95% at 323 K, which aligns closely with experimental observations. Concurrently, our calculations show that the relative energy of the transition state leading to linear product formation (TS-Linear) is comparable to that of TS-branched-*S* ($\Delta\Delta G$ = 1.0 kcal/mol), which is consistent with experimental findings and further consolidates that *b*/*l* regioselective control is very difficult. Further analysis of relevant transition state structures reveals that in TS-Linear, during S−C bond formation, there has been a significant transformation from an η^3-Pd structure to an π-Pd structure. The conjugation between the carbon−carbon double bond in the linear product and its aromatic group contributes to enhance electronic stability, explaining why arylallene typically favors linear product generation as noted in previous studies. Although steric hindrance exists between phosphorus ligands on the catalyst and two aromatic groups along with their functional groups within TS-Linear, rendering it less favorable compared to TS-branched-*S*, a comprehensive consideration of both electronic effects and steric hindrance indicates that their final energy difference remains relatively small. In addition, we are pleased to report that 1,1-substituted allene substrates can also produce highly enantioenriched branched products with the Pd-**L18** chiral metal complex, thereby providing an effective approach for constructing chiral quaternary carbon centers. The corresponding transition state structures are also calculated, and the computational results show that the relative energy difference between TS-branched-*S*-Me and TS-branched-*R*-Me has not decreased (more details shown in SI), thus maintaining a high stereochemical selectivity.

In summary, we have developed a palladium-catalyzed/ligand-controlled highly regioselective asymmetric hydrosulfonylation of simple allenes. Using different diphosphine ligands, Josiphos or Segphos, the reaction of alkylallenes or arylallenes with sulfinic acids afforded the corresponding chiral branched allylsulfones in high yields and high enantioselectivities, respectively. The known Pd-catalyzed hydrofuncationalization of allenes generally gives the linear allyl compounds; this reaction herein breaks through this intrinsic mode, and the described protocol has the advantages of good atomic economy and operational simplicity as well as good functional tolerance. And it was noteworthy that the chiral allylsulfones with a chiral quaternary carbon stereocenter could be also generated with high enantioselectivities, as was comparable to that in Aponick's work on the asymmetric hydrosulfonylation reaction of allenes; only one chiral allylsulfone with a quaternary carbon stereocenter was obtained in low enantioselectivity. In addition to the experimental results on

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the ligand-controlled regioselectivity and enantioselectivity, the DFT calculations also consolidated the great challenge of the regioselectivity and enantioselectivity control for the arylallenes in detail. This is the first application of Pd-catalyzed/ligand-controlled asymmetric hydrosulfonylation of simple allenes, and further studies on the other asymmetric functionalization of allenes are underway in our laboratory to expand this protocol of Pd-catalyzed processes.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.4c04259.

The characterization data, X-ray single crystal diffraction data of **3oa** (2327860) and **3sa** (2327861), spectra of all products (^{1}H NMR, ^{13}C NMR, ^{19}F NMR and HPLC) and the DFT calculations detail (PDF)

FAIR data, including the primary NMR FID files, for compounds **3aa**−**3bba** (ZIP)

Accession Codes

Deposition Numbers 2327860−2327861 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Author Contributions

All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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